

Can Repair Arise from Changed Relational Organization Rather Than an Explicit Stored Target?

A minimal non-neural carrier experiment

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Materials: github.com/AmirTlinov/relational-closure-carrier

BIOLOGICAL BOUNDARY

Work on anatomical homeostasis and basal cognition treats morphogenesis as a form of collective problem-solving in morphospace: a body can reach or recover large-scale organization despite perturbation, while physiological networks retain information that is not reducible to current anatomy alone [1–4]. I am exploring a narrow mechanistic question inside that program:

Can target-like repair arise from history retained in the relational organization of the same substrate that later conducts repair, without a separate complete-body target model, global controller, reward signal, or online external selection inside the modeled loop?

The motivating hypothesis is relational:

No relation is physically meaningful unless its sides are distinguishable.

Отношение физически содержательно лишь постольку, поскольку его стороны различимы.

Motion or contact alone need not create an operative unit. When distinguishable states modify one another and remain coupled so that their relation changes later propagation, they form what I call a **held difference**: a group that can act as one operative relational unit without erasing its internal distinction. At the scale of a body, injury then changes not only a local site but the organization of the surviving whole. Repair need not reproduce the previous component exactly; it may restore a group-level capability by forming a different relation with the surviving material.

This is an experimental heuristic, not a demonstrated account of biological regeneration or a claim about fundamental physics. The computational experiment asks only whether one clean, interventionally testable fragment of this hypothesis can be realized.

The proposed connection to relativity is programmatic: physical description may be grounded in relations between distinguishable sides rather than privileged absolute labels. No result in relativity is derived here. In particular, the tested invariance under hostile address relabeling is invariance to graph storage labels, not a spacetime transformation, Lorentz invariance, or general covariance.

MINIMAL CARRIER

The prepared arena is a deterministic, non-neural computational dynamical system over explicit relations. The harness-designated persistent surface contains 1,024 writable relational sites and is called **BODY** for protocol provenance. Each site stores a small local state and explicit handles to available contacts and conducting relations. One finite local collision rule is used throughout. It has no global sweep, coordinate arithmetic, fitness function, symbolic planner, runtime comparison with a completed form, or channel carrying reward.

The same rule can:

1. conduct change through an existing relation;
2. occupy material available through the current relations;
3. retract a route that fails to close;
4. form a higher-order conducting relation only after a lower route physically returns to its origin.

The external harness prepares the arena, partitions storage, and reads the result, but it does not select a relation during execution. The local law has no predicate for that witness partition.

The temporary five-record surface labeled **WORLD** is another file-backed region of the same **MaterialRecord** physics. One ongoing continuation follows explicit handles across both witness partitions. A return is counted as earned only if that continuation first changes the particular surrounding record from which the return later emerges:

ongoing continuation → changed surrounding record → earned return → closure

Passive contact, a disconnected surrounding change, or a prewritten return state therefore cannot substitute for this material provenance sequence.

WITNESS BOUNDARY AND SURFACE FLATTENING

BODY and **WORLD** are harness/witness labels for two mmap storage surfaces governed by the same **MaterialRecord** format and local collision physics. This partition belongs to the experimental architecture: the surface router uses its namespace to choose a file, while the frozen **collision** law contains no **BODY**/**WORLD** predicate and follows only explicit handles.

A dedicated control asked whether that routing namespace is required for the local dynamics of one prepared matched passage. The witness copied all 1,024 former **BODY** records and five former **WORLD** records into one direct 1,029-record **MmapBody**, removed the surface namespace from every non-sentinel handle, and applied a deterministic hostile fixed-point-free permutation to all nodes. The ordinary two-surface run and the flat run were then compared through the same frozen collision kernel.

All 2,058 / 2,058 collision events were isomorphic in full—kind, current place, met place, continuation, and changed-place tuple—and all 1,029 / 1,029 final **MaterialRecord** values were isomorphic. Thus the **BODY**/**WORLD** namespace is not dynamically necessary for this matched trajectory. The witness still constructs the mapping and remembers the former labels.

HISTORY-DEPENDENT REPAIR ASSAY

Two lines, A and B, begin from the same injured 513-site **BODY**-labeled snapshot. Their **WORLD** fixtures have equal resources and differ only in the physical placement of the earned return into one of two prepared routes in that persistent group. Both lines recover the full 1,024-site group and the same higher-order conducting capacity.

The harness then deletes the temporary **WORLD** files. Only each persistent **BODY**-labeled snapshot survives. Every snapshot is reopened in a fresh process, exposed to the same common washout, and cut again at the same current relation. Future repair is probed with fresh A and B fixtures in both orders (AB and BA).

The question is not whether a log can classify the histories. It is whether the surviving placement of relations within the persistent group is necessary for different future conduct.

RESULT

Across seeds 17, 23, 41, and 59, histories A and B produced persistent groups with the same material extent and the same immediately available large-scale conductance, but with different placements of return relations earned during passage through records labeled **WORLD**. These differences survived deletion of the original temporary file, restart of the persistent snapshot, common washout, and a second injury.

After that injury, line A restored the higher-order relation on the A opportunity and line B on the B opportunity. Reversing probe order changed the episode number but not the line-specific response. Within-line signatures were identical across seeds; A/B signatures remained separated.

The result was tested over two generations with seven paired controls per line and seed, for 112 comparisons. Controls removed the **WORLD** fixture, made contact passive, disconnected change of the surrounding record from the ongoing continuation, mismatched the continuation phase, delayed return under the same finite collision budget, redirected the return to the other prepared route, or replaced continuation-dependent formation in **WORLD**-labeled records with a pre-conducting phase-incompatible route. All 112 controls separated from the matched condition both in the resulting **BODY** bytes and in behavior after a further common washout and injury.

Two hostile interventions localize the memory claim:

- a complete relabeling of storage addresses, while preserving relations, preserved both **BODY**-labeled relational organization and future behavior;
- shuffling only the placement of the history-bearing return relations erased the A/B future difference while preserving the amount of material and the rest of the relational structure.

Together these interventions define **Relational Distinguishability** in this carrier: changing storage labels preserves the A/B distinction, while changing the distribution of the candidate return-relation class removes it. That distribution is interventionally necessary for the observed A/B difference; the result does not localize memory to a small site.

A separate scale intervention removed only the higher-order conducting relation. The large-scale act disappeared while the lower route remained. A new pass through that surviving route restored the higher-order relation.

The bounded result is therefore:

In this prepared carrier, two histories of contact across witness-labeled partitions are retained as different relational placements within the persistent group; they remain distinguishable in later repair after temporary-file deletion, restart, common washout, and renewed injury, and the distribution of the tested return-relation class is interventionally necessary for that distinction.

The current source passes 22 tests; the public evidence bundle records four seeds, 112 paired controls, and the single-mmap isomorphism receipt. World-lineage executable source SHA-256:
d3ed7438372973ef17b61b01659569372bb8c9faf61c48b334ded6bedca962e8.

LIMITS AND THE QUESTION I AM ASKING

The topology and witness partition are engineered. The experimenter specifies the available contacts, the higher-order closure affordance, the two return routes, the injury, and the readout. No organism/environment or **BODY**/**WORLD** predicate is available to the local law. Calling the held **BODY**-labeled relational group an organism is a proposed interpretation: in this assay it remains a witness-selected group among other records of the same physics, not a demonstrated self-born boundary. The carrier does not discover an open-ended morphology, maintain metabolism, or establish its own goals. It is digital, uses a small state vocabulary, and is not a model of ion channels, gap junctions, or biological tissue. It has not been independently replicated. I do **not** present it as evidence of basal cognition or as an explanation of morphogenesis.

The flattening result is deliberately narrower than the serial assay. It covers one prepared matched passage only; deletion of the temporary **WORLD** file, restart of the persistent **BODY**-labeled snapshot, common washout, G2, and the paired control family have not been reproduced in the flat field. It therefore does not show a self-born boundary or establish that every use of the **BODY**/**WORLD** witness partition can be eliminated from the full protocol.

Its intended value is narrower: it probes the boundary between **history-dependent self-organization** and **anatomical homeostasis**. The runtime contains no separate symbolic target register or model of a completed body. However, the history-bearing relational placement is readable through future repair and rewritable by intervention, while the prepared higher-order slot and return routes encode a closure affordance. Together, these may already constitute a rudimentary distributed pattern memory or homeostatic target rather than its absence.

My main question is:

Would you regard the topology-encoded closure affordance and history-bearing relational state as a minimal distributed pattern memory or homeostatic target, or only as a path-dependent attractor? What single perturbation would best distinguish those interpretations?

REFERENCES

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